

Experience:

Data Engineer

GROPYUS GmbH, Germany

May 2022 – Present

- Developing Data-fabric using knowledge graphs to enable next-gen data driven smart buildings
- Developed a tool to generate a RDF knowledge graphs from various data sources in just few seconds using Python & RDFLib
- Used unit-testing using Pytest and SHACL shapes with Pyshacl to validate graphs
- Worked on a research product which estimates global warming potential, carbon emission and other sustainability KPIs of our buildings using Okobaudät dataset
- Deployed various applications and services on Azure using Kubernetes.
- Developed a tool using OpenAI GPT and Langchain to talk with RDF KG and generate SPARQL queries

Student Research Assistant

DICE Research Group, Germany

July 2020 – September 2021

- Lead development of a chatbot (IDA) which let you upload datasets and generate visualization using natural language
- Helped in project management, development of Natural language engine and dockerized the project and created CI/CD pipelines using Jenkins

Software Engineer

Syqlo.de, Germany

June 2018 – June 2020

- Developed a web-based manager for a house connection surveying app using Angular and Firebase

Software Developer Engineer

Symantec Inc, Pakistan

November 2016 – March 2018

- Worked on a real-time dashboard using tools including Angular 1, Sockets, Python's Django and MongoDB

NodeJS Developer

Interactive Cells, Pakistan

March 2016 – October 2016

- Worked with NodeJS & Cassandra DB to develop REST APIs and migration scripts for an Ad-tech startup

Projects:

Let Me help

Hackathon project

- A hackathon project which let user to control computer using natural language instruction. I team up with a friend and we implemented SOTA research papers and were **finalist!** Later Claude introduce [same functionality](#). I also wrote detailed blog [post](#). [github.com/Stettiner-Valley/LetMeHelp](#)

RDF-to-Text (NLG task)

Master's thesis (grade 1.5)

- Researched Knowledge graphs (RDF)-to-text generation using *Generative Adversarial Networks (GANs)*
- Developed my solution in a pipeline fashion (5 stages) using Pytorch and WebNLG corpus
- Used LSTM as a generator, CNN model as a discriminator, and Monte-Carlo as a Policy Gradient method
- Trained and evaluated 3 neural networks using SLURM workload manager on UPB's Nvidia GPUs clusters

IDA

Master's project (grade 1.3)

- A chatbot for data-scientist, Co-lead the team of 8 fellow students and developed visualizations and Natural Language Engine. [github.com/dice-group/ida-pg](#)

Classifier using Transformer's Encoder

Personal project

- As Transformer is a sequence-to-sequence model so was curious to explore how to use Encoder separately! Developed only using Pytorch. [github.com/maqbookhan/Transformer_classifier_pytorch](#) (6 Github ★)

Project NMT

- A project where I explored LSTM and Transformer from scratch, pre-trained word embeddings and Huggingface models for Neural Machine Translation using Pytorch. [github.com/maqbookhan/nmt](#)

Project NLP

Personal project

- A project where I implemented basic tasks of NLP e.g., Classification using statistical and deep learning approaches. github.com/maqboolkhan/project-NLP

A Web Search Engine

Bachelor's final year project

- A search engine focused on query understanding and grouping relevant results using Naïve Bayes. Developed using NLTK, Flask, Nodejs and Elasticsearch DB.

Oranje

The Simplest Java Code Editor. oranjehq.blogspot.com

ACCS

A Rapid Microsoft word document format changer.

maqhq.blogspot.com

Skills:

Programming Languages

Python

Java

Javascript

Python Frameworks

Pytorch

Scikit-Learn

Numpy

Pandas

Langchain

RDFLib

PySHACL

FastAPI

PyTest

ML/NLP

Transformer

BERT

LSTM

NMT

Attention

Word embeddings

Sequence to Sequence models

Huggingface

Databases

Elasticsearch

Triple stores & MongoDB

Vector

SQL

SPARQL

stores

Tools

Docker

Git

Regex

Kubernetes

Jupyter Notebooks

CI/CD

Education:

Masters in Computer Science (Major in AI) Paderborn University, Germany Mar 2018 - Sept 2021

Bachelors in Computer Engineering Sir Syed University of Engineering, Pakistan Jan 2012 - Dec 2015

Extra Circular Activities:

- Student at ACL Mentorship program (Aug 2021)
- First Pakistani Winner of Firefox OS Global App Dev Competition (Nov 2014)
- Third Position in two days Scayle Hackathon (Nov 2016)